Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahaj-bhadja sahaj-bhadja

Summary

Software engineer with **2+ years of experience** building **C++/Python-backed systems**, performance-sensitive pipelines, and cross-platform tooling, with strong alignment to **optimization**, **benchmarking**, **Linux-style engineering workflows**, and ML-adjacent systems software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Architected a **multi-tenant AI platform** on Azure SQL and Blob Storage for **500+ users**, reducing dashboard latency from **1.2s to 350ms** through query tuning, indexing, and systems-level performance optimization.
- Built **FastAPI** execution pipelines across PostgreSQL, SQL Server, and cloud storage, processing **5K+ monthly jobs** with validation, debugging, and release checks that sustained **99%+ reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed and tested a **Flask + React** grading platform serving **150+ instructors**, handling **10K+ submissions** with concurrent processing, secure ingestion, and production-grade backend workflows.
- Designed multithreaded backend services on AWS RDS with REST APIs and **CI/CD**, sustaining **10K+ requests/day** while reducing feedback turnaround by **60%** through profiling and pipeline optimization.

USC AD Labs **Los Angeles, California**
ML Research Assistant *January 2025 – May 2025*
- Built simulation and model-training pipelines for autonomous-driving planning, processing **100K+ trajectory logs** and improving long-horizon planning quality by **18%** across closed-loop evaluation workloads.
- Architected an HD-map-aware spatial-temporal encoder in **PyTorch**, improving closed-loop stability by **15%** while working on performance-sensitive research software for robotics-style systems.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservice software with **Jenkins**, PyTest, and Mockito, automating build, test, and delivery flows across **3 operating systems** with **20+ unit/integration suites**.
- Optimized API and ETL workflows using **AWS Glue**, **S3**, and OOP patterns, improving response time by **30%**, boosting pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Pandit Deendayal Energy University **India**
Research Fellowship *January 2023 – May 2024*
- Formulated a quantum-resistant blockchain design by integrating **5+ lattice-based cryptographic primitives**, strengthening distributed-system resilience and algorithmic rigor in research software design.
- Analyzed consensus mechanisms and developed a novel algorithm that reduced centralized-control and quantum-risk exposure by **25%** in benchmark simulations, contributing to a peer-reviewed paper presented at IISc Bengaluru.

Skills

Languages: Python, Java, C++, JavaScript/TypeScript, SQL
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, Microservices
Frontend: React.js, Next.js, TailwindCSS
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL
Cloud & Infrastructure: AWS (S3, RDS), Microsoft Azure, Docker, Kubernetes, Jenkins, GitHub Actions, Linux
AI/ML: PyTorch, Transformers, NLP, LLM Systems, Reinforcement Learning
Engineering Practices: CI/CD, Unit & Integration Testing, Distributed Systems, Performance Optimization, Benchmarking, Technical Documentation, Agile

Projects

CPU-scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*
- Implemented **6 CPU scheduling algorithms** and an interactive Gantt-visualization workflow, evaluating **1K+ simulated process combinations** and reducing debugging time for OS-style scheduling logic by **20%**.

Chat2SQL Live | *Java, TypeScript, SQL, Docker*
- Worked across an **open-source** Java and TypeScript codebase to support **10+ databases**, cross-platform builds, and AI-assisted query workflows using Docker, Maven, and Node.js tooling.

CaptionCraft Live | *Transformers, PyTorch, Flask*
- Benchmarked transformer architectures on a **10K image / 100K caption** dataset, improving caption accuracy by **15%** while building inference APIs for ML-framework-backed model serving.